

A Digital Platform for Corporate Carbon Emission Accounting, Analytics, and ESG Reporting

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Literature Review

Carbon emission accounting and reporting have gained significant attention over the past two decades due to the escalating urgency of climate change and the global push toward sustainability. Organizations across sectors are increasingly expected to measure, manage, and disclose their greenhouse gas (GHG) emissions as part of environmental, social, and governance (ESG) responsibilities. Carbon accounting has therefore evolved from a voluntary sustainability initiative into a strategic and regulatory necessity. The literature on carbon accounting highlights the growing importance of accurate measurement, transparency, and standardized reporting, while also identifying major challenges in methodology, comparability, and practical implementation.

Early studies on carbon footprint estimation emphasize the complexity of defining and measuring emissions. Researchers note that there is often inconsistency in system boundaries, emission sources, and the types of gases included in calculations. Some methodologies focus only on carbon dioxide, while others include methane, nitrous oxide, and additional greenhouse gases expressed in carbon dioxide equivalents. This lack of uniformity has historically made it difficult to compare results across organizations or industries. Scholars argue that without standardized frameworks, carbon footprint assessments risk becoming unreliable and potentially misleading. These foundational concerns established the need for globally recognized standards and accounting protocols.

The development of formal frameworks such as the Greenhouse Gas Protocol and ISO-based guidelines marked a major milestone in the field. These frameworks introduced structured classifications of emissions into Scope 1 (direct emissions), Scope 2 (indirect emissions from purchased energy), and Scope 3 (value chain emissions). Literature consistently recognizes these frameworks as the backbone of modern corporate carbon reporting. However, several studies point out that the flexibility allowed within these standards can also create inconsistencies. Organizations may choose different consolidation approaches, operational boundaries, or emission factors, which can significantly alter reported results. As a result, comparability across firms remains a persistent challenge.

A recurring theme in academic research is the voluntary nature of many carbon disclosures. While reporting has increased, the quality and depth of disclosures vary widely. Some organizations provide detailed, data-driven reports, while others present high-level estimates with limited methodological transparency. Researchers suggest that voluntary reporting may encourage selective disclosure or “symbolic” sustainability actions rather than substantive emissions reductions. This has led to calls for stronger regulatory oversight and standardized digital reporting systems that reduce subjectivity and improve data reliability.

Another important area of research focuses on the integration of carbon accounting into managerial decision-making. Traditionally, carbon reporting was treated as a compliance or corporate social responsibility activity. However, recent literature highlights the strategic value of emissions data. When integrated into management control systems, carbon accounting can inform operational efficiency, energy use optimization, supply chain decisions, and long-term investment strategies. Despite this potential, studies indicate that many firms still struggle to embed carbon data into core business processes. Emissions data often remain siloed within sustainability departments rather than being used as a decision-support tool across the organization.

Life Cycle Assessment (LCA) has emerged as one of the most rigorous methods for evaluating emissions across product and process life cycles. Academic research strongly supports LCA as a scientifically robust approach, particularly for industries such as manufacturing, transportation, and construction. LCA enables organizations to account for emissions from raw material extraction to end-of-life disposal. However, literature also highlights limitations, including data intensity, cost, and the need for specialized expertise. Public ESG reports often simplify or partially apply LCA methods, which can reduce accuracy and credibility. Researchers suggest that digital platforms could automate and

standardize LCA-based calculations, making them more accessible for organizations.

Scope 3 emissions receive significant attention in recent literature because they often represent the largest share of a company's carbon footprint. These emissions arise from upstream and downstream activities such as supplier operations, logistics, product use, and disposal. Scholars consistently note that Scope 3 measurement is highly complex due to fragmented data sources and reliance on estimates. Many companies underreport or exclude Scope 3 emissions due to methodological uncertainty. Benchmarking studies show that industry-wide averages are often used in place of primary data. This indicates a major opportunity for digital tools that can streamline Scope 3 data collection and estimation.

Another critical issue identified in research is the lack of harmonization in emission factors and global warming potential (GWP) values. Different reporting standards and jurisdictions may use different time horizons or conversion factors when translating non-CO₂ gases into CO₂ equivalents. Studies demonstrate that such differences can create substantial reporting gaps, particularly for methane emissions. This variability complicates efforts to track progress toward climate targets. Scholars therefore emphasize the importance of methodological transparency and standardized factor databases within carbon accounting systems.

Recent literature also explores the role of digital technologies in carbon tracking. Advances in data analytics, artificial intelligence, blockchain, and Internet of Things (IoT) sensors are reshaping how emissions data are collected and processed. Smart meters, automated data pipelines, and AI-driven analytics can improve accuracy and enable near real-time monitoring. However, these technologies introduce new challenges related to data governance and privacy. Research indicates that high-precision carbon tracking systems may require sensitive operational and financial data. This creates a trade-off between measurement accuracy and privacy protection. Organizations must therefore balance transparency with data security.

Dashboards and visualization tools are widely discussed as effective mechanisms for communicating sustainability data. Business intelligence research shows that dashboards enhance decision speed, pattern recognition, and performance monitoring. Visual analytics can translate complex emissions datasets into actionable insights for managers and stakeholders. Scholars emphasize that well-designed dashboards support key performance indicator tracking, scenario analysis, and forecasting. In the context of carbon accounting, dashboards can

make emissions data more accessible and useful for non-specialist users, thereby strengthening organizational engagement with sustainability goals.

Interoperability and system integration are additional themes in recent studies. Carbon data often originate from multiple sources, including energy management systems, enterprise resource planning platforms, and supply chain databases. Literature highlights the difficulty of consolidating these data streams into a unified reporting system. Lack of interoperability can lead to duplication, errors, and reporting delays. Researchers advocate for integrated digital platforms that can aggregate data across systems while maintaining consistency and auditability.

The literature also points to growing regulatory pressure as a driver of carbon reporting improvements. Governments and international bodies are increasingly mandating sustainability disclosures and climate-related risk reporting. This shift is pushing companies toward more robust and transparent accounting practices. However, regulatory requirements vary by region, which can complicate multinational reporting. Scholars suggest that standardized digital tools could help organizations comply with multiple frameworks simultaneously by mapping data to different reporting standards.

Finally, academic research underscores the importance of credibility and auditability in carbon reporting. Investors and regulators are placing greater emphasis on verified data. Without reliable systems, companies risk reputational damage and regulatory scrutiny. Digital platforms that maintain traceable data records, calculation logs, and downloadable reports can enhance trust and accountability. Researchers view such systems as essential for the future of corporate sustainability reporting.

Overall, the literature reveals a clear trajectory: carbon accounting is transitioning from fragmented, manual reporting toward standardized, technology-driven systems. Yet significant gaps remain in standardization, Scope 3 measurement, interoperability, and data governance. These gaps highlight the need for integrated digital platforms that can automate calculations, provide real-time analytics, ensure methodological transparency, and support regulatory compliance. Such platforms have the potential to transform carbon accounting from a reporting obligation into a strategic management tool that drives meaningful emissions reductions.

Literature Table

Title	Journal / Source	Key Findings	Relevance to Carbon Emission Tracking Platform
A Systematic Review of Corporate Carbon Accounting and Disclosure Practices: Charting the Path to Carbon Neutrality	<i>Journal of Research in Emerging Markets</i>	Carbon disclosures are rising but inconsistent, voluntary, and often lack technical detail.	Shows need for standardized digital reporting platforms to ensure consistency and comparability.
The Effectiveness of Carbon Accounting in Reducing Corporate Carbon Footprints	<i>International Journal of Multidisciplinary Research and Growth Evaluation</i>	Carbon accounting is effective when integrated with management systems and transparency; Scope 3 remains difficult.	Supports dashboard systems that identify emission hotspots and integrate with management decisions.
Carbon Accounting: Challenges for Research in Management Control and Performance Measurement	<i>Abacus</i>	Carbon accounting is not well integrated into management control and performance systems.	Indicates need for tools that connect emissions data to business decision-making.
From Measurement to Monitoring: Evaluating Privacy Risks in Carbon Footprint Tools	<i>IEEE Access</i>	High-accuracy tools collect sensitive data; strong privacy–accuracy trade-offs exist.	Highlights importance of privacy safeguards and secure system architecture.
Sustainability Accounting for GHG Measurement Using the GREET LCA Model	<i>Clean Technologies and Environmental Policy</i>	LCA improves accuracy but ESG disclosures often lack verifiable data.	Encourages integration of LCA-based calculation modules.
Carbon Footprint: Current Methods of Estimation	<i>Environmental Monitoring and Assessment</i>	No coherence in footprint definitions, gases, or boundaries across studies.	Emphasizes need for standardized calculation methodologies.

Title	Journal / Source	Key Findings	Relevance to Carbon Emission Tracking Platform
Lack of Harmonisation of GHG Reporting Standards and the Methane Emissions Gap	<i>Nature Communications</i>	Different GWP metrics create large reporting gaps.	Platforms should standardize emission factors and document methods.
Accounting for Carbon Emissions—Current State under the GHG Protocol	<i>Sustainability</i>	Firms mostly use operational control consolidation, reducing comparability.	Platforms can standardize consolidation methods for fairness.
Bibliometric and Systematic Review of Carbon Footprint Tracking in Cross-Sector Industries	<i>Sustainability</i>	Key gaps: lack of standards, poor interoperability, limited real-time tracking.	Supports integrated, cross-sector tracking platforms.
The Role of Dashboards in Business Decision Making and Performance Management	<i>Anvesh Doctoral Research Conference in Management</i>	Dashboards improve decision speed and monitoring effectiveness.	Strong justification for visual analytics and dashboards.
Unpacking Carbon Accounting Numbers: Commensurability and Comparability of GHG Disclosures	<i>Journal of Cleaner Production</i>	Facility-level reporting is more comparable than corporate aggregates.	Suggests including facility-level tracking modules.
Category-Specific Benchmarking of Scope 3 Emissions for Corporate Clusters	<i>Renewable and Sustainable Energy Reviews</i>	Scope 3 dominates emissions but requires complex data collection.	Opportunity for Scope 3 estimation tools.
Harmonizing Global Carbon Accounting Standards	<i>International Journal of Research in Marketing Management and Sales</i>	GHG Protocol, ISO, and regulatory frameworks differ significantly.	Platforms can act as harmonization layers.
Developing Robust Accounting Models for Quantifying Scope 3 Emissions	<i>International Journal of Financial Management and Economics</i>	Hybrid LCA and probabilistic models improve accuracy.	Advanced analytics can enhance precision.

Title	Journal / Source	Key Findings	Relevance to Carbon Emission Tracking Platform
Creative Accounting in Scope 2 Emissions Reporting	<i>Energy Policy</i>	Market-based methods can misrepresent renewable energy impact.	Platforms should include transparency and reporting notes.

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